

M-66901



Newport®

Oriel Product Line

150 Long Beach Boulevard

Stratford, CT 06615-0872

Phone: (203) 377-8282

(800) 714-5393

Fax: (203) 378-2457

E-MAIL oriel.sales@newport.com

**UNIVERSAL ARC LAMP HOUSINGS
500 WATT FAMILY**

USER MANUAL

Please read these instructions completely before operating this equipment. The specification and operating instructions apply only to the model(s) covered by this manual. If there are any questions or problems regarding the use of this equipment, please contact Newport or the representative from whom this equipment was purchased.

Rev: 06-10-08

TABLE OF CONTENTS

I. INTRODUCTION	1
II. SAFETY CONSIDERATIONS	3
II.1 RADIATION	3
II.2 LAMP EXPLOSION	3
II.3 OZONE	4
II.4 ELECTRICAL SHOCK	4
II.5 EMI	5
II.6 HEAT	5
III. GENERAL DESCRIPTION	6
III.1 LAMP AND REFLECTOR ADJUSTMENTS	6
III.2 BUILT-IN IGNITOR	6
III.3 LAMP COOLING	6
III.4 SAFETY AND MONITORING FEATURES	6
III.5 HOUSINGS WITHOUT CONDENSING LENSES	7
III.6 MOUNTING	7
IV. LAMP INSTALLATION & INITIAL OPERATION	10
IV.1 INSTALLATION OF THE LAMP	10
IV.2 ADJUSTMENT OF LAMP, MIRROR AND LENS	13
IV.2.1 GENERAL	13
IV.2.2 ADJUSTMENTS PRIOR TO OPERATION	13
IV.2.3 ADJUSTMENTS DURING OPERATION	13
IV.2.4 LAMP OPERATION AND COOLING	14
IV.3 COOLING	15
V. LAMP HOUSING OPTICS	16
V.1 COLLIMATED BEAMS	17
V.2 UNIFORMITY	18
V.3 IMAGING THE ARC	19
V.4 REAL LENSES	19
V.4.1 SPHERICAL ABBERATIONS	20
V.4.2 CHROMATIC ABBERATIONS	21
V.5 POSITIONING THE CONDENSER TO GET A COLLIMATED BEAM	21
VI. TROUBLESHOOTING	22
VI.1 INTRODUCTION	22
VI.2 PROBLEMS	22
VI.3 INPUT CONNECTOR	26
VII. DECLARATION OF CONFORMITY	27
VIII. WARRANTY AND RETURNS	28

LIST OF FIGURES

Figure 1: Lamp And Reflector Adjustments.....	8
Figure 2: Dimensional Diagram.....	9
Figure 3: Correct operating position for Arc Lamps	11
Figure 4: Typical lamp adapter configurations	12
Figure 5: Electrodes inverted view	13
Figure 6: Correctly positioned reflector overlays the inverted arc image on the arc gap.....	14
Figure 7: Collimated Beams	17
Figure 8: The collimated output of a 200W Hg lamp in the vertical and horizontal planes.	18
Figure 9: A source focused to a smaller image.	19
Figure 10: Spherical Abberations	20
Figure 11: Chromatic Aberration	21
Figure 12: Starter wire installation on Xe lamp.	23
Figure 13: Lamp Housing Connector Pin Assignment	26

I. INTRODUCTION

This manual covers these Oriel 50 to 500 watt(W) Universal Arc Lamp Housings with internal ignitors:

66901	500W Lamp Housing with F/1.5	UV grade fused silica condenser
66902	500W Lamp Housing with F/1	UV grade fused silica condenser
66903	500W Lamp Housing with F/0.85	Pyrex [®] condenser
66904	500W Lamp Housing with F/0.7	Aspherab [®] silica-borosilicate crown condenser
66905	500W Lamp Housing with F/0.7	Aspherab [®] UV grade fused silica condenser
66912	250W Lamp Housing with F/0.7	Aspherab [®] borosilicate crown condenser
66900	500W Lamp Housing without condenser (not CE compliant)	
67001	500W Lamp Housing with F/2.2	UV grade fused silica collimating condenser
67003	500W Lamp Housing with F/2.2	UV grade fused silica focusing condenser
67005	500W Lamp Housing with F/1	UV grade fused silica collimating condenser
67006	500W Lamp Housing with F/1	UV grade fused silica focusing condenser

The housings have the same basic design and differ in the condensing lens assembly. With the appropriate socket adapter you can interchange between Hg, Xe and Hg(Xe) lamps of different power. An ignitor is built into the top of the lamp housing. This reduces RF interference and eliminates the need for a separate ignitor.

NOTE: The optics for the Model 66912 are rated only to 250W. Do not use higher powered lamps in this housing.

These lamp housings are designed to operate with Oriel Arc Lamp Power Supplies, Models 68905, 68907, 68910, 68911, 69907, 69910 and 69911, which provide all necessary power and control. Table 1 on the next page shows which power supply you should use and the appropriate socket adapter for each lamp.

These lamp housings, when used in conjunction with the appropriate Oriel power supply, are compliant with the European EMC and Low Voltage Directories and, therefore, carry the CE mark.

Table 1: Appropriate Socket Adapters and Power Supplies for Oriel 500W Universal Lamp Housings

Lamp			Power Supply	
Type	Model No.	Socket Adapter	Type	Model No.
50 W Hg	6282	66158	50-200 W Universal	68907, 69907
75 W Xe	6251	66150	50-200 W Universal	68907, 69907
75 W Xe OF	6263	66150	50-200 W Universal	68907, 69907
100 W Hg	6281	66150	50-200 W Universal	68907, 69907
100 W Xe OF	6257	66150	50-200 W Universal	68907, 69907
150 W Xe	6253	66151	50-200 W Universal	68907, 69907
150 W Xe UV	6254	66151	50-200 W Universal	68907, 69907
150 W Xe OF	6255	66151	50-200 W Universal	68907, 69907
150 W Xe compact	6256	66152	50-200 W Universal	68907, 69907
200 W Hg	6283	66144	50-200 W Universal	68907, 69907
			200-500 W Hg	68910, 69910
200 W Hg(Xe)	6289	66169	50-200 W Universal	68907, 69907
			200-500 W Xe/Hg(Xe)	68911, 69911
200 W Hg(Xe) OF	6290	66169	50-200 W Universal	68907, 69907
			200-500 W Xe/Hg(Xe)	68911, 69911
*300 W Xe	6259	66160	200-500 W Xe/Hg(Xe)	68911, 69911
*300 W Xe OF	6258	66160	200-500 W Xe/Hg(Xe)	68911, 69911
*350 W Hg	6286	66161		68910, 69910
*400 W Xe	6260	Not Required	200-500 W Xe/Hg(Xe)	68911, 69911
*500 W Hg	6285	66162	200-500 W Hg	68910, 69910
*500 W Hg(Xe)	66142	66159	200-500 W Xe/Hg(Xe)	68911, 69911

* Lamps above 200W must not be used in the 66912 housing.

II. SAFETY CONSIDERATIONS

SUMMARY OF HAZARDS

There are six hazards in the operation of systems employing these lamp housings. They are:

- Radiation
- Lamp Explosion
- Ozone
- Electrical Shock
- EMI
- Heat

The interlock system is designed for your safety. Do not defeat the interlocks.

II.1 RADIATION

The high intensity UV radiation of these lamps can permanently damage the cornea, lens, and retina of the eye, even causing blindness. This damage may not be immediately apparent. The deep UV is absorbed in the cornea or eye fluids; focused UV, VIS and NIR can damage the retina. Normal blink reaction to visible light may not be adequate protection, and a beam of invisible UV or NIR (produced by spectral filtering) can be most dangerous as the blink response is not induced. UV radiation can also cause painful sunburn, and with prolonged exposure, serious burns.

Recommendations

- Never look directly into the output beam of a housing when operating a lamp.
- Never look at a specular (mirror) reflection of the beam, even for short periods of time.
- Always wear UV safety eyewear or face mask, and protective clothing for exposed areas of skin.

II.2 LAMP EXPLOSION

When cold, xenon and mercury (xenon) arc lamps are under several atmospheres of pressure and are subject to explosion due to internal strains or to physical abuse. When hot, all lamps are under a pressure of close to 100 atmospheres and subject to violent explosion.

Recommendations

- Do not handle a bare arc lamp without safety goggles and adequate protection for exposed areas of skin.
- Do not apply torque to the lamp envelope during installation or removal.
- Do not touch the lamp envelope. Fingerprints and other contaminations left on the lamp cause a deterioration of the envelope during operation and may lead to lamp explosion.
- Check the lamp polarity before starting the lamp.
- Thoroughly clean the envelope with alcohol or a dilute solution of detergent and water after installation in the housing.
- Attach a cover plate over the condenser aperture of any housing that will be used without the

condenser in place before operation.



II.3 OZONE

Shortwave ultraviolet light photolyses oxygen to produce ozone. This is emitted in the cooling air stream of the lamp housing. There is no simple way of predicting the ozone concentration (or its impact on you) e.g., operation in a small enclosed area may lead to high concentrations. Operation of the same system in a large, well ventilated laboratory may not be a problem. Recommended maximum exposures are typically:

0.1 ppm for 8 hours exposure
2 ppm for a 2 hour exposure

A very sensitive nose can detect 0.015 ppm. 1 ppm produces a strong and obnoxious odor. As a rule of thumb, if you can easily smell ozone, the level is too high for prolonged exposure.

Recommendations

- Use an ozone free lamp unless you need the shortwave UV.
- Operate the system in a large ventilated area.
- Use an Oriel Ozone Eater™.

II.4 ELECTRICAL SHOCK

When this lamp housing is operated as intended, with an Oriel Model 68905, 68907, 68910 and 68910 Power Supply and Oriel Model 70050 or 70051 cable, the interlock system and the package design eliminate risk of electrical shock.

Use of this lamp housing with a power supply not manufactured by Oriel is not recommended.

Be aware that a high transitory voltage is used to ignite the lamp and, before ignition, the lamp terminals have a potential difference of up to 200 V. These voltages are dangerous and it is wise always to approach the lamp housing as if the interlock system were not in place.

Recommendations

- Keep personnel clear of all exposed terminals.
- Before changing lamps or working on the system, check the power supply voltmeter to be sure that internal capacitors are discharged.
- Make sure all connections are securely made and check the polarity before starting a lamp.
- Do not handle the lamp leads during lamp ignition.
- Attach a cover plate over the condenser aperture of any housing that will be used without the condenser in place, before operation.

II.5 EMI



This lamp housing with its associated cables and an Oriel Model 68905, 68907, 68910 and 68910 Arc Lamp Power Supply contains EMI (Electro Magnetic Interference) under normal operating conditions.

However, ignition of an arc lamp requires high voltage pulses to break the lamp down and a high current dump (kHz discharge) to sustain the arc. The ignition creates significant electromagnetic energy. The high level of EMI integrity of the source contains most of this transient energy, but good earthing, cable routing practice, and EMI shielding may be necessary to protect sensitive digital circuitry from these pulses.

Recommendations

- Start the arc lamp before powering nearby computer systems.
- Keep the computer at least 6 feet away from the ignitor/power supply.
- Use a different outlet and line for the computer and ignitor/power supply.

II.6 HEAT



These arc lamps become very hot after several minutes of operation, and remain in such state for many minutes after being shut off.

The Oriel Arc Lamp Power Supplies will continue to operate the lamp housing fan to cool the lamp off after the lamp/power supply has been shut off. As long as the power supply is connected to the mains the fan will run until the lamp has cooled down.

Recommendations

- Use the power supply circuit breaker/power switch to turn the lamp off. Do not use a switched mains receptacle since this will prevent the fan assisted lamp cool down.
- Wait for the housing fan to stop, or at least 10 minutes after turning off the lamp, before opening the lamp housing door.
- Approach the lamp as if it were hot under any circumstances.

III. GENERAL DESCRIPTION

These lamp housings are lightweight but rugged. They are excellent choices for most low to medium power laboratory lamps. You can mount them to optical tables, rails, or benches. Various models with different condensers are available. With the complete line of optical accessories described in Volume II, you can filter, focus, and steer the output to fit your application.

III.1 LAMP AND REFLECTOR ADJUSTMENTS

The position of the arc (relative to the terminals which secure the lamp) varies slightly from lamp to lamp. This is because of normal arc lamp manufacturing tolerances. These lamp housings have adjustments to place the arc of a new lamp in the same position as that of the lamp being replaced. Three fine controls position the reflector. Then, independent controls position the arc horizontally and vertically. The reflector and its controls travel with the vertical and horizontal controls. This eliminates the need for iterative lamp-reflector adjustments. See Figure 1 for location of adjusters.

III.2 BUILT-IN IGNITOR

The housings described in this manual have an ignitor built into the top of the housing. Short arc lamps require a high voltage spike, up to 40 kV for Xe lamps, to ignite. Having the ignitor built into the lamp housing minimizes unwanted radiated or conducted EMI and reduces potential exposure to the high voltage ignition energy.

III.3 LAMP COOLING

The built in fan and housing baffles maintain the proper operating temperature for arc lamps, when operated in normal laboratory ambient. Lamp housing is designed to operate in a typical laboratory environment (typically 68 to 76 degrees F, up to 45% relative humidity). Temperature and humidity outside of typical laboratory range can contribute to cooling and ignition faults. Cooling issues which will cause the over temperature sensor to open and ignition problems from high humidity. Contact Newport representative for more information if operating outside the suggested range.

Overheating due to blocking of the cooling vents or an inoperative fan activates a thermostat interlock which disables the power supply.

The fan speed is controlled to keep the lamp at a constant operating temperature. The fan starts slowly when the lamp is ignited, speeds up as the lamp approaches the operating temperature, runs at (almost) constant speed after the lamp has warmed up and continues to run after the lamp has been turned off until the lamp has cooled down.

The cooling fan receives its power directly from the 68905, 68907, 68910 or 68910 power supply.

NOTE: The fan cooling will not be effective if the condenser is removed and the aperture is not covered.

III.4 SAFETY AND MONITORING FEATURES

These lamp housings incorporate safety interlocks. When used with an Oriel Power Supply the lamp automatically shuts off if the housing door is opened or the housing overheats. If you use your own power supply, we strongly recommend that you utilize this low voltage interlock system for safety.

III.5 HOUSINGS WITHOUT CONDENSING LENSES

If the condenser lens is removed, there will be a large aperture in the front of the housing. This opening compromises the integrity of the housing and exposes the operator to the potential hazards of electrical shock and lamp explosion and affects the lamp cooling process. The condenser aperture **must** be securely covered before operating the lamp.

III.6 MOUNTING

The lamp housings are supplied with four adjustable feet for free standing use. They allow 0.63 inch (16 mm) height adjustment. You can also mount the housing to tables, carriers, or translators with tapped holes on 1 inch or 25 mm centers. To do this you should remove the feet and use the three mounting points and screw slots. If you have a flat enough surface you can unscrew the three mounting pads and use the baseplate directly. See Figure 2 for baseplate dimensions.

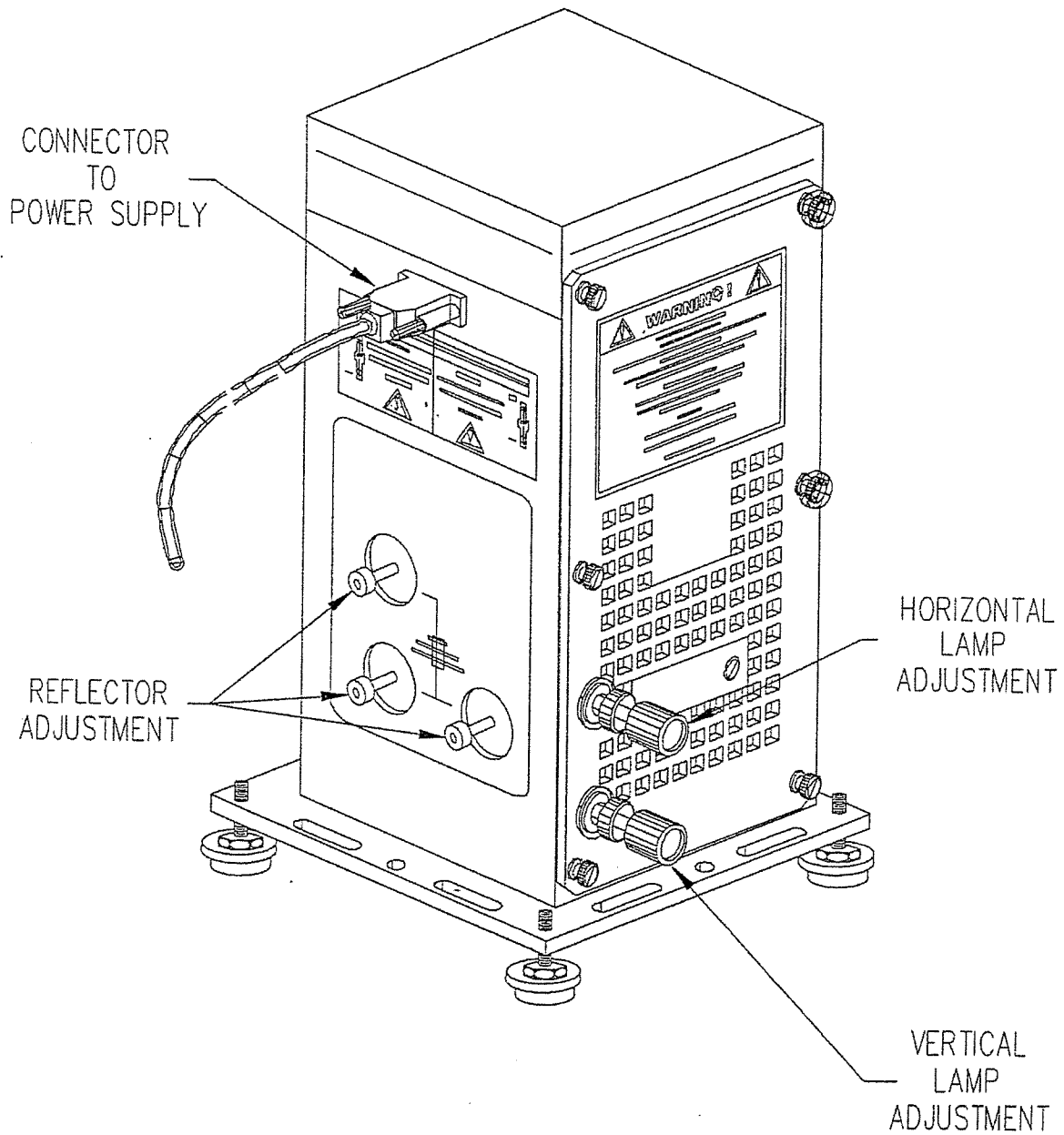


Figure 1: Lamp And Reflector Adjustments

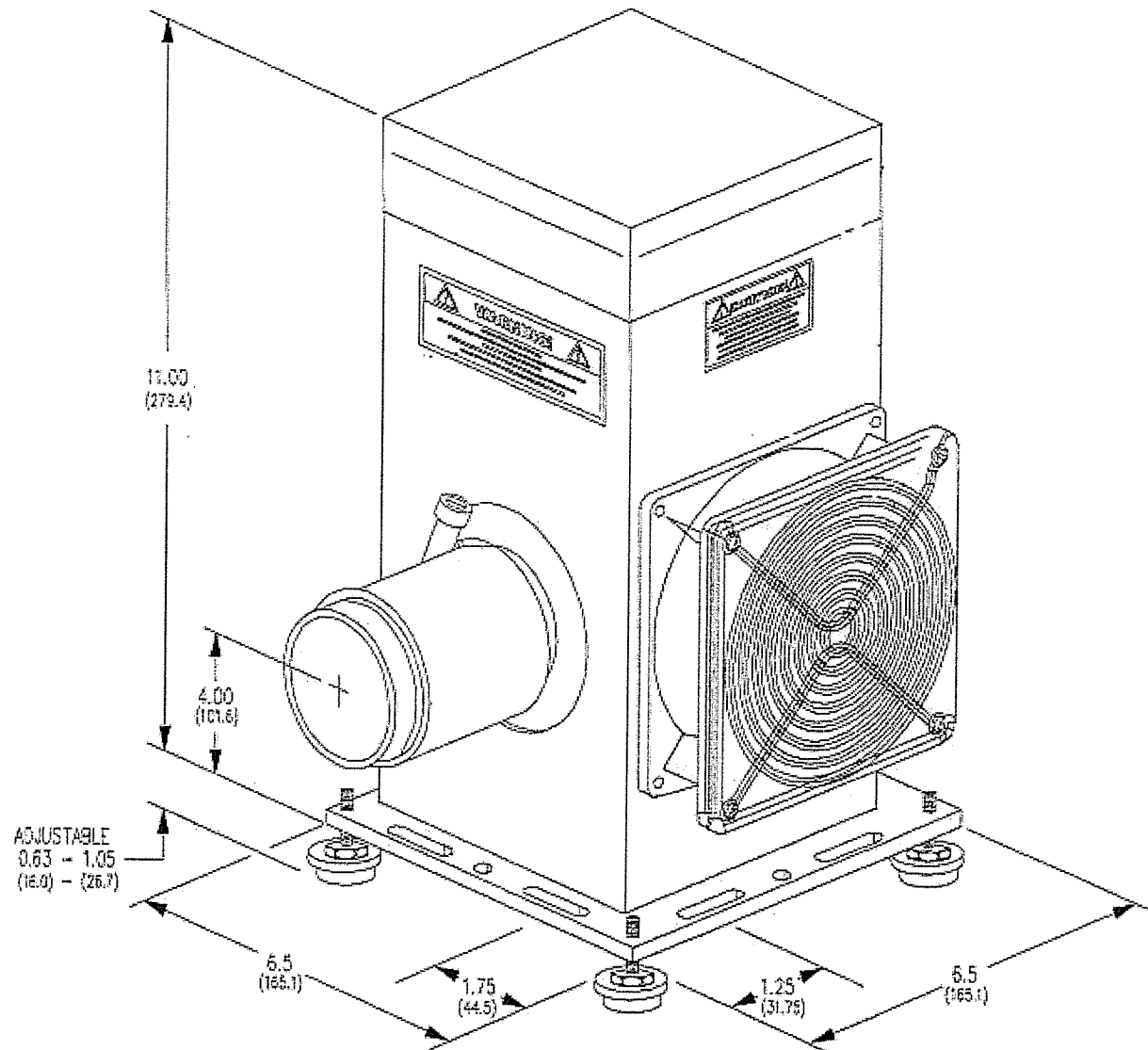


Figure 2: Dimensional Diagram

IV. LAMP INSTALLATION & INITIAL OPERATION

IV.1 INSTALLATION OF THE LAMP

To install the lamp:

- Check that there is no electrical service to the housing and that the lamp power supply is off and discharged.
- Put on safety goggles and gloves.
- Unscrew the four thumb screws on the side of the housing and remove the access door.
- Determine which orientation is correct for the lamp. For xenon and mercury (xenon) lamps, the anode (+ve) is at the top. For mercury lamps the anode (+ve) must be at the bottom. You can identify the anode by a + stamped on the lamp base. Any lamp identification writing or numbers will be the correct way up when the lamp is held in the correct operating orientation. See Figure 3 for a diagram.
- Install the appropriate adapter(s) on the lamp terminal(s). Depending on the lamp type, these socket adapters are fastened to the lamp terminals by set screws or by a threaded connection. Refer to Figure 4 for typical adapter configurations.
- If the lamp is supplied with knurled nuts, remove the bottom nut and discard before installing the socket adapter. One of the knurled nuts will be used to secure the sensor block assembly to the top adapter.
- Connect the sensor block assembly on the top hanging lead to the top adapter, using either the knurled nut supplied with the lamp, or the hardware supplied with the adapter. Refer to Figure 4 for typical connection information.
- Set the lamp position adjusters at approximately mid range. Move the adjusters while watching the bottom mounting block to find mid range. Place the bottom terminal (with adapter) into the bottom mounting block. Adjust the lamp vertically so the arc gap lies approximately in the center of the mirror, then tighten the knurled set screw on the side of the mounting block.
- After the lamp is in position, clean the envelope with alcohol and lint free tissue. Fingerprints left on the lamp may cause the lamp to explode when lit.
- Replace the side access door and tighten the knurled thumb screws.
- Connect the lamp cable between the lamp housing and the rear of the power supply. Use the black cable, **Model 70050**, to run xenon and mercury (xenon) lamps; the gray cable, **Model 70051**, for mercury lamps.
- Read the Power Supply Instruction Manual before operation.

<<< CAUTION >>>

BE SURE THE LAMP IS POSITIONED WITH THE PROPER ELECTRODE ON TOP AND THE CORRECT CABLE CONNECTING THE POWER SUPPLY AND THE LAMP HOUSING. OPERATION WITH THE WRONG POLARITY WILL IMMEDIATELY DESTROY THE LAMP.

It is important to re-check the lamp polarity. If you are operating the lamp housing for the first time or changing lamp type, **DO IT NOW**. It may save you a lamp. Incorrect polarity will result in rapid destruction of the lamp.

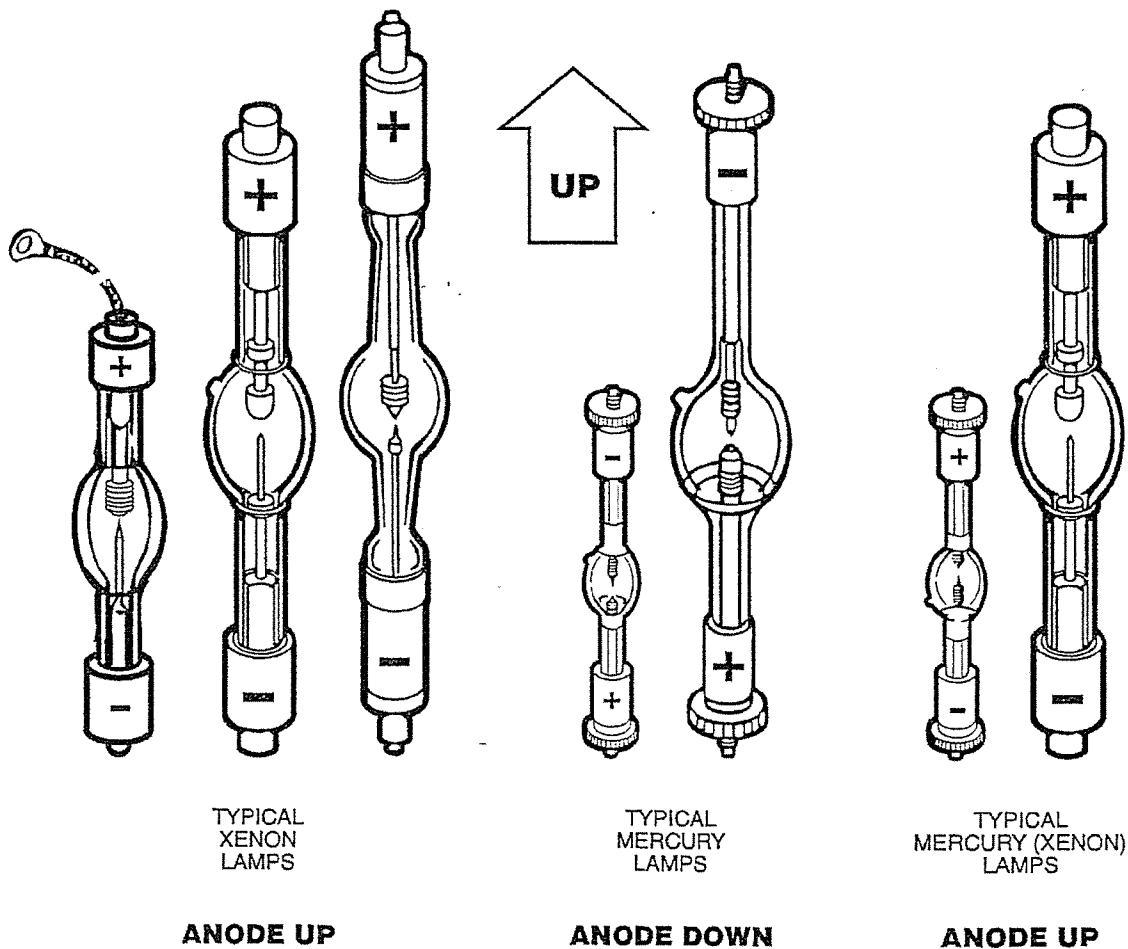


Figure 3: Correct operating position for Arc Lamps

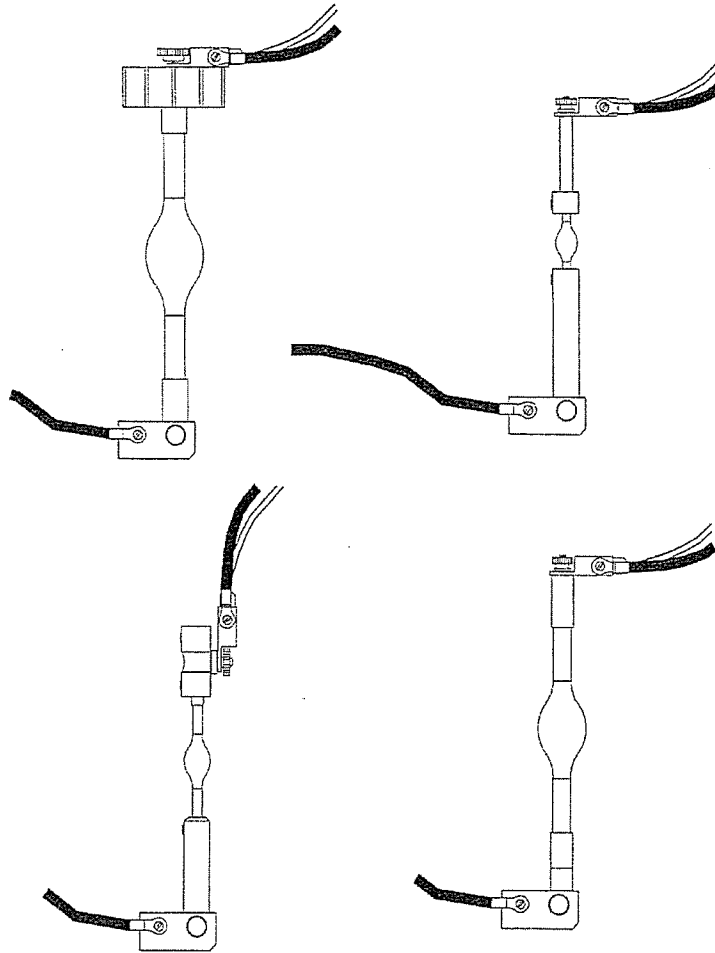


Figure 4: Typical lamp adapter configurations

IV.2 ADJUSTMENT OF LAMP, MIRROR AND LENS

IV.2.1 GENERAL

In these housings, the spherical mirror behind the lamp creates an inverted arc image, which can be placed onto or near the lamp arc to increase the output beam intensity. The condensing lens collects radiation from both the arc and the mirror image of the arc.

The mirror is adjusted with respect to the lamp by three knobs (smaller) located directly behind the mirror on the back of the housing. The lamp and mirror in unison are adjusted by a vertical and horizontal adjust knob (larger) in the same area. See Figure 1.

<<< CAUTION >>>
DO NOT LET THE MIRROR IMAGE OF THE ARC FALL ONTO EITHER ELECTRODE. OVERHEATING OF THE LAMP SEALS AND SUBSEQUENT LAMP EXPLOSION MAY RESULT.

IV.2.2 ADJUSTMENTS PRIOR TO OPERATION

1. Before ignition, roughly adjust the lamp position with the horizontal and vertical adjustment knobs. Adjust the lamp so the arc gap lies approximately in the center of the condensing lens. This can be seen through the lens if the focusing lever on the lens is full back.
2. If you have appropriate safety equipment, ignite and warm up the arc lamp per the instructions in the power supply manual.

IV.2.3 ADJUSTMENTS DURING OPERATION

1. By adjusting the condensing lens, focus the output beam so an image of the arc appears on a wall or screen. Do not view UV images without safety glasses.
2. Adjust the lamp position to center the output in the condenser lens aperture. You can place a piece of paper over the condenser output and center the beam on the aperture.
3. Rotate the mirror knobs until a bright spot appears alongside the arc (See Fig. 5). This bright spot is a distorted image of the arc.

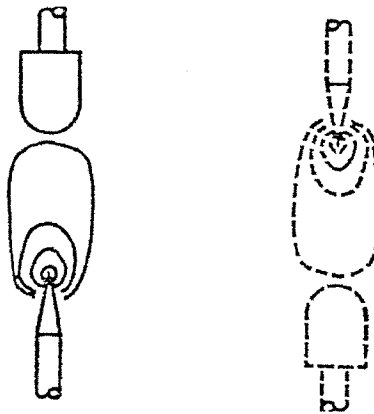


Figure 5: Electrodes inverted view

4. Focus the image by rotating all three screws in the same direction concurrently (this translates the mirror back or forth). When focused you will see an inverted image of the arc, about the same size as the arc.
5. Place the mirror image over the main (direct) image (as in Fig. 6) or alongside, as desired. You may need to use iterative adjustments to keep a focused mirror image.

<<< CAUTION >>>

YOU CAN SUPERIMPOSE THE TWO IMAGES, BUT AVOID REIMAGING THE ARC HOT SPOTS ONTO THE ELECTRODES. THIS CAN CAUSE OVERHEATING OF THE LAMP.



Figure 6: Correctly positioned reflector overlays the inverted arc image on the arc gap

6. Focus the condensing lens as desired.
- If additional range of adjustment of the condensing lens is desired:
1. Remove the condensing lens assembly handle (take care to record the orderly assembly of its parts).
 2. By hand, slide the inner lens barrel forward or back until another tapped hole appears in the spiral slot.
 3. Reassemble the handle into this new hole.

IV.2.4 LAMP OPERATION AND COOLING

These arc lamps should be operated close to their rated power. Dropping the power below 80% of rated, can lead to unusual lamp performance, eventual instability, and shortened life. With mercury lamps it can also lead to cooling problems and unstable light output.

The electrodes as viewed on a screen in front of the condenser lens are inverted (right). The reflector image as a screen (left) is doubly inverted so the anode of a xenon or mercury (xenon) lamp appears on top.

Correctly positioned reflector overlays the inverted arc image on the arc gap. The image is the same size as the arc itself.

IV.3 COOLING

These housings are fan cooled. The fan ensures that the lamp terminals are at an appropriate temperature and the lamp housing skin temperature is safe. The fan speed in these housings is thermally controlled so that the lamp top electrode is kept at a constant temperature as room temperature and lamp cooling requirements vary.

The fan control cycles through four phases:

lamp warm up phase: after ignition the fan remains off as the lamp warms up

fan ramp up phase: when the lamp reaches a preset temperature the fan comes on at low speed and speeds up as necessary as the lamp continues to warm up until the lamp temperature reaches the control set point

temperature control phase: the fan speed is modulated to keep the temperature at the set point

cool down phase: when the lamp is turned off, the fan comes on at full speed to cool down the lamp as quickly as possible (as long as the power supply is plugged into the mains voltage supply). The fan will shut off when the lamp has been cooled.

This constant temperature operation minimizes the concern about over or under cooling the lamp unless there are other cooling mechanisms being used - most likely ducting and/or other blowers that are used to remove the ozone by-product that can either augment or restrict the normal cooling air. Before hooking up ducting or other blowers, note the operating voltage and current of the lamp. Check these parameters after adding to your system.

If too much cooling air is provided, the lamp will be overcooled. An overcooled lamp never reaches proper operating conditions and will run with low voltage / high current and often unstable light output. The high current operation will shorten the lamp life. Mercury lamps are very sensitive to overcooling as the mercury does not vaporize.

The evidence of an under-cooled lamp is less obvious. The lamp will warm up quickly, but the voltage and current will be similar to a normal lamp. Since excessive temperature can damage the lamp seals and lead to premature - sometimes dramatic - failure, care has to be taken if the air flow around the housing is changed significantly.

A thermal interlock in these housings will activate and turn off the power supply in a grossly undercooled condition. In the event of a failure in the temperature control circuitry, the fan should run at full speed providing overcooled but safe operation.

V. LAMP HOUSING OPTICS

Different types of condenser lenses are available on these Lamp housings. You can check the lens type on your housing from the listing:

66901	500W Lamp Housing with F/1.5 UV grade fused silica condenser
66902	500W Lamp Housing with F/1 UV grade fused silica condenser
66903	500W Lamp Housing with F/0.85 Pyrex [®] condenser
66904	500W Lamp Housing with F/0.7 Aspherab [®] silica-borosilicate crown condenser
66905	500W Lamp Housing with F/0.7 Aspherab [®] UV grade fused silica condenser
66912	250W Lamp Housing with F/0.7 Aspherab [®] borosilicate crown condenser
67001	500W Lamp Housing with F/2.2 UV grade fused silica collimating condenser
67003	500W Lamp Housing with F/2.2 UV grade fused silica focusing condenser
67005	500W Lamp Housing with F/1 UV grade fused silica collimating condenser
67006	500W Lamp Housing with F/1 UV grade fused silica focusing condenser

The lenses were designed for efficient collection of light from the arc. In order to get the best performance from your Lamp housing, we first review some aspects of light collection and then describe how to set the lens position.

By moving the focusing lever you can move the position of the condenser lenses to produce a diverging beam, "collimated beam" or to re-image the arc. The lenses in these housing are designed for collimation rather than imaging. The lens shape and orientation are selected to minimize lens induced distortions (aberration) when the lenses are close to the position which produces a collimated beam (the collimating position). When you use them for imaging, there are two penalties; lens aberrations increase* and light collection is reduced. For imaging, the lens is moved further from the arc, and so gathers less of the light emitted by the arc within its aperture. The lens operates at a higher F/#.

***See Volume III for a comprehensive discussion on aberrations.**

If you need to image the arc close to the Lamp Housing, or equivalently, produce a small image of the arc, then it is more efficient to use the condenser in the collimating position and use a secondary focusing lens to create the image.

To simplify the discussion, we first describe the operation of an ideal lens and then some of the major results of aberrations.

V.1 COLLIMATED BEAMS

The usual concept of a collimated beam is a parallel cylinder of light. If the intensity is the same anywhere across a section of the cylinder, the beam is uniform. Unfortunately there is no source of a uniform, perfectly collimated beam. Even expensive laser sources have some residual divergence, in the limit governed by the laws of diffraction, and they usually have non-uniform, though sometimes known, intensity distributions.

Arc lamp sources with an ideal condenser lens in the collimating position produce beams, which depend on the source size and intensity distribution.

A pinhole source at the focus of an ideal lens produces a beam, which is close to the ideal collimated beam. In Figure 7, we show a second pinhole source a distance "d" from the first.

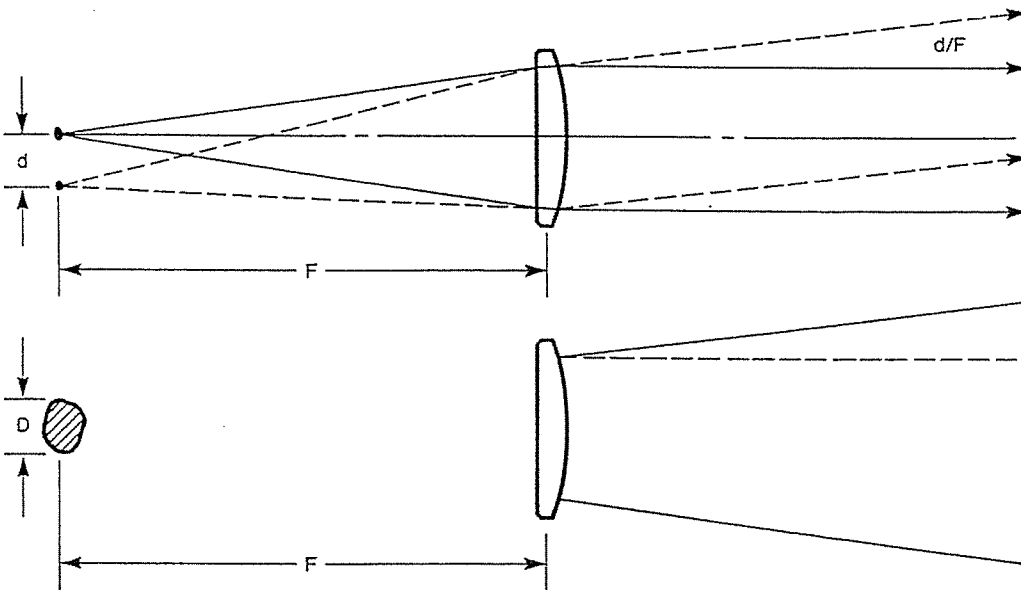


Figure 7: Collimated Beams

For most arc sources the divergence in one plane is not the same as that in the orthogonal plane.

The beam from the second pinhole is collimated, but at an angle $\arctan(d/F)$ with the first. Any extended source can be thought of as a whole set of touching pinholes. The beam after the lens is the sum of all the beams from all the pinholes. It will contain rays with angles up to $\arctan(D/F)$ where D is the largest dimension of the source. The beam will have a divergence, which depends on the sum of the light from all the points on the source. Obviously this divergence will depend on the size of the source and the intensity of the various "pinholes" or points on the source.

Most arc sources are non-uniform and are not circular. Therefore the divergence in one plane is not the same as that in the orthogonal plane. For most design purposes the arc sizes quoted in Volume II and the lens focal length give a good guide to divergence. For low divergence consider the small arc sources and, if necessary, our Spatial Filter Assembly (Volume II).

V.2 UNIFORMITY

Arc sources are not uniform and usually have an intensity peak near one electrode. Intensity contours shown in Volume II indicate arc uniformity. The collimated beam comes from pinhole sources, which are not equally intense in addition to being spatially distributed. The result is a smooth, non-uniform beam with some divergence. Figure 8 shows a scan of a collimated beam in the vertical and horizontal planes. The source was a 200W Hg arc lamp.

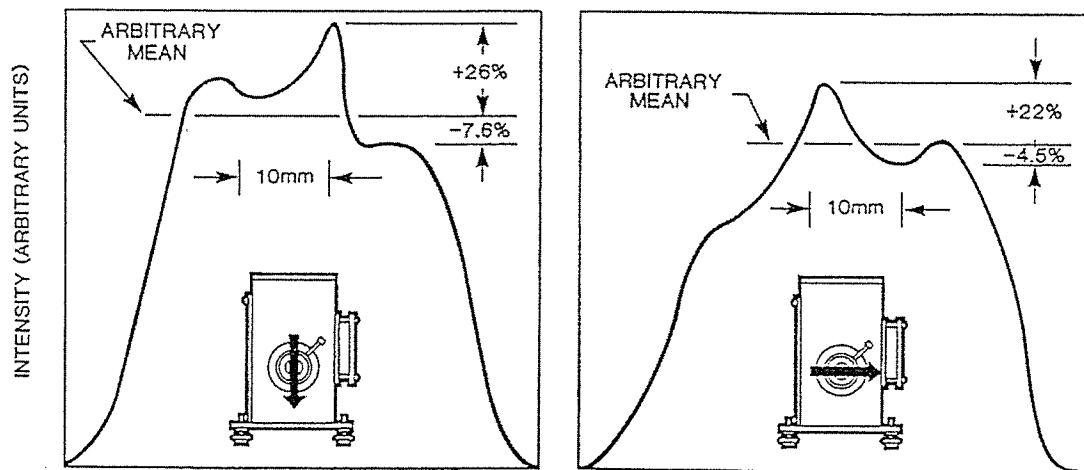


Figure 8: The collimated output of a 200W Hg lamp in the vertical and horizontal planes.

V.3 IMAGING THE ARC

You can reimage the arc by positioning the condenser further from the arc using the focusing lever. Volume III describes imaging and provides the formulae. As the condenser lens is moved out, the image moves in and becomes smaller. As already indicated, the lens collects less light as it is moved away from the arc. Additionally, the convergence angle of the beam goes up as the image becomes smaller. This is not usually important for irradiance of a surface, but can be significant if the image is on the slit of a monochromator, optical fiber, or other optical system with limited acceptance angle. We normally use a secondary focusing lens to maximize the light through a slit or into a fiber optic. Figure 9 shows the higher convergence produced when creating a small arc image.

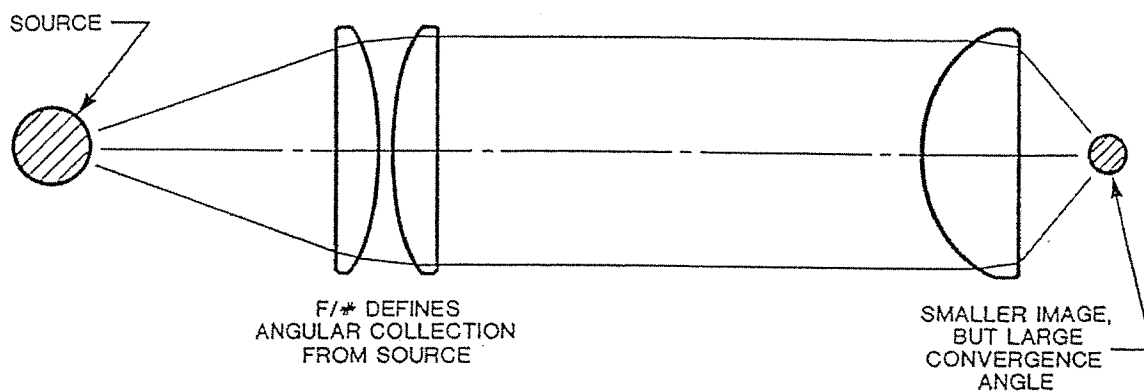


Figure 9: A source focused to a smaller image.

V.4 REAL LENSES

The condenser lenses are intended for efficient light collection. They operate at low F/#s. As a result the single element F/0.85 and F/1 lenses suffer from severe aberrations, particularly spherical aberration. The doublet F/1.5 lens is somewhat better, while the four element Aspherabs™ are almost free from spherical aberration. Note that all the lenses perform best while collimating the light from the source.

V.4.1 SPHERICAL ABBERATIONS

This aberration results from the fact that the ideal lens, the aberration free lens, is not spherical in shape. With the exception of the aspheric condenser, these condenser lenses, like most lenses, have spherical surface shapes for economic manufacturing. In general, spherical aberration is decreased by dividing the refraction (light bending), as equally as possible between as many surfaces as possible. The lens shapes (plano-convex for the fused silica singlets) of our condensers and orientations minimize spherical aberration for the type of condenser and at the collimating position. The F/0.7 Aspherab™ multi-element lens assembly practically eliminates this aberration by balancing the effects between lenses.

Consider the simple plano convex lens collimating light from an ideal point source. With the plano surface towards the point source and the point source at the (paraxial) focus the marginal rays converge while the paraxial rays are collimated (Fig. 10). This is due to spherical aberration. For the ideal, non-spherical lens shape, the paraxial and marginal rays are all collimated. If the source is located about $1/4 f$ inside the focus, the paraxial rays diverge slightly and the marginal rays are almost collimated. This is often the optimum compromise for a single element collimating lens (and has the added advantage of collecting more light from the source).

The lens adjustment on these lamp housings allows the lens to be moved closer to the source than the paraxial focus. You can empirically find the best position for your system.

The F/0.7 Aspherab™ condensing lenses combine negligible spherical aberration and low F/#, and should be used at focus for critical collimating applications.

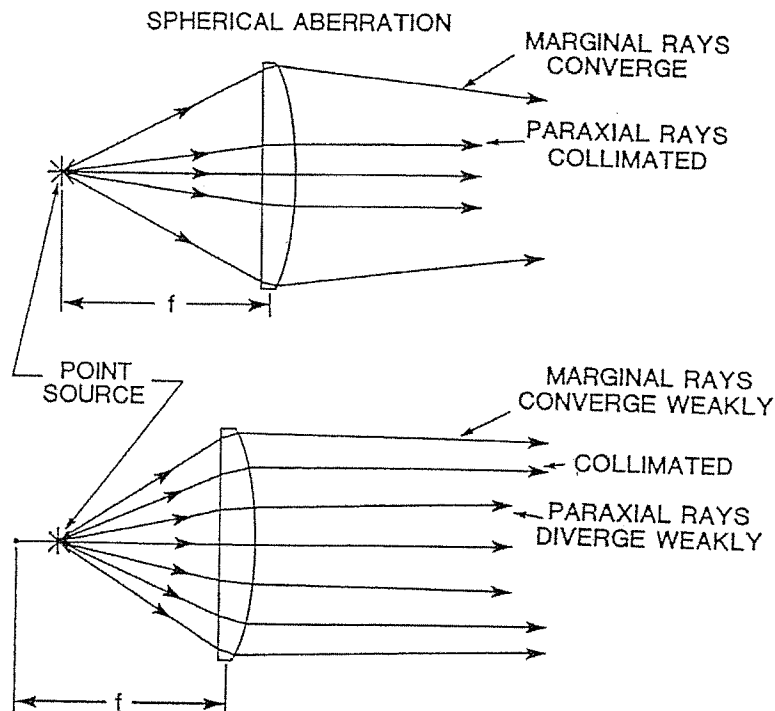


Figure 10: Spherical Abberations

The marginal rays and the paraxial rays are collimated when the point source is at the paraxial focus, and the plano surface of the lens faces the point source.

V.4.2 CHROMATIC ABBERATIONS

The term "chromatic aberration" describes the variation of lens focal length with color (Fig. 11). This variation is due to the change in the lens index of refraction (n) with wavelength. As the wavelength goes up, n goes down and the focal length increases.

This causes problems in producing multi-wavelength collimated beams, but is usually a second order effect compared with source and spherical aberration limitations. Chromatic aberration usually becomes significant only when deep UV wavelengths are collimated. The refraction in f (focal length) for a fused silica lens from the visible value of f to $0.91 f$ at 250 nm may require a change of lens for optimum performance. Contact Oriel for details.

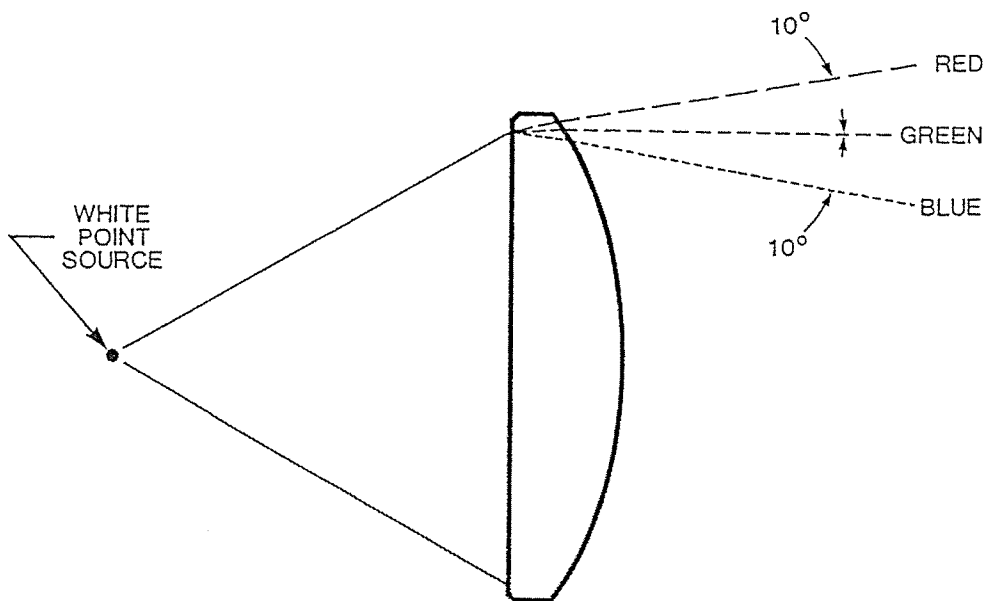


Figure 11: Chromatic Aberration

Chromatic Aberration: different wavelengths are focused at different points.

V.5 POSITIONING THE CONDENSER TO GET A COLLIMATED BEAM

You should image the arc on the most distant wall in your laboratory (remembering appropriate safety measures) to get close to the collimation position. You can then move the lens barrel in a small amount for best collimation. If your wall is 2 meters away, a 1.3 mm adjustment is required. For 3 m and 4 m, the corresponding numbers are 0.8 and 0.6 mm.

VI. TROUBLESHOOTING

VI.1 INTRODUCTION

This section deals with procedures to follow if you encounter specific difficulties in operating these Lamp Housings. Additional details for repairing the Lamp Housing circuitry are available from Oriel, but in general we do not advise attempting to work on the circuitry as this could expose you and your equipment to the high ignition voltage.

VI.2 PROBLEMS

Lamp will not light after several repeated presses of the LAMP START button.

The most common problem experienced when using these Lamp Housings is difficulty in starting the lamp. The problem may be in the Lamp Housing, in the Power Supply, or with the lamp. The following procedure should help you identify the problem area. If you cannot locate the source of the problem, and do not have other Lamp Housings or Power Supplies to interchange as a problem finding technique, we recommend you send the complete system, Power Supply, Lamp Housing and lamp to Oriel for diagnosis of the failure mode.

Recommended Procedures

1. Check that the Power Supply is operating -visually inspect the power breaker and make sure the fan is operating.
2. 699xx Power Supplies:
Check for fault codes on front panel of power supply. Recommended correctives action is contained in the power supply manual.

699xx Power Supplies:

Check that there is pre-ignition voltage available from the Power Supply. Move the toggle switch on the Power Supply to the voltage position and check that there is an open circuit voltage of at least 60 volts. If not, check the interlock circuit. If there is no pre-ignition voltage then the interlock circuit may not be closed. The principal interlock is the lamp housing door so check that it is closed. There are safety thermostats located in both the lamp housing and the power supply that will break the interlock circuit if a fan fails. It is very unlikely that either thermostat is activated.

If you cannot trace the fault, contact Newport for advice. Do not operate the Lamp Housing with a defeated interlock.

3. Check the connection between the Lamp Housing and Power Supply. Make sure the cable is firmly attached and the correct cable for your lamp.
4. Check the lamp and the internal connection to the lamp.

Remove the lamp cable from the Housing, open the Lamp Housing and check that the lamp connections are tight. You may need to remove the lamp and examine it. Assuming no catastrophic damage, then check for cracks in the lamp or lamp stem. Examine the molybdenum strip conductors inside the lamp stem for continuity. Small breaks in these conductors will prevent lamp operation. Examine the electrodes for excessive "burn back" or rounding. If the lamp has a trigger wire, then check that it is properly attached. If it is a xenon or mercury(xenon) lamp, and it does not have a trigger wire, and you would like to attach one, follow this procedure:

- Use a high temperature nichrome wire, AWG 28, with an OD of 0.315 mm.
- Wearing cots or gloves loop the wire around the top of the lamp envelope, (A) in Fig. 12, twisting the end securely.
- Bring the wire down the side of the envelope, (B), nearest to the vacuum "fill/break off" mark. Keep the wire snug against the glass.
- Loop the wire around the snug of the envelope (C).
- Continue this wire, keeping it snug against the glass, down the side of the envelope. Wrap tightly around bottom ferrule and twist the end to ensure it does not slide.

5. Check for ignition.

If all the connections are properly made and the lamps seems fine, then place a piece of paper near the output of the condenser lens and if possible darken the room. Use safety goggles and view the paper while pressing the LAMP START button. What you see will help isolate the problem.

A dim flash, and brief snapping noise indicates that the ignitor is breaking down the lamp, but that the main capacitor in the power supply is not dumping. This indicates that the power supply ignitor drive circuit is working but there is inadequate break down, due either to a problem with the DC output section of the power supply, or a lamp problem.

A brighter flash indicates that the ignitor is breaking down the lamp and the power supply is dumping the output capacitor, but that the power supply is failing to sustain the discharge. The ignitor is operating properly, the lamp or power supply are suspect.

No flash, and no "snap" at all indicates the ignitor or the power supply ignitor circuitry is not working. The ignitor itself, the cabling to the power supply, or the drive circuitry in the Power Supply is at fault.

Depending on the test results, either the Lamp Housing or the Power Supply should be returned to Newport for repair or replacement.

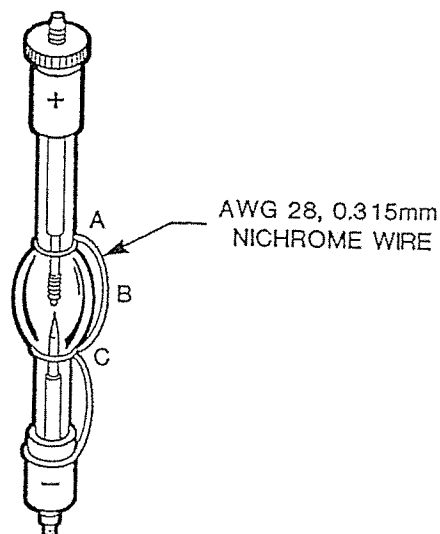


Figure 12: Starter wire installation on Xe lamp.

Lamp Housing Fan does not operate properly.

The Lamp Housing Fan is driven by the lamp's operating temperature and does not start immediately when power is applied. The fan will run until the lamp has cooled off as long as the power supply is plugged in. In most cases, running the Lamp Housing without fan operation can be dangerous. Any malfunction of the fan circuit should be remedied immediately.

Recommended procedure

1. Check that the Lamp Housing cable is connected correctly and that the power supply is connected to an active mains and/or is turned on. If the fan still fails to operate properly, contact Newport for further assistance.

Mercury lamps do not reach the correct current/voltage.

If a mercury lamp is not running at the correct temperature, it will not reach the correct current/voltage. If the lamp is running at its rated power but the lamp current is too high and the voltage is too low, the lamp is being overcooled. If the lamp is running at its rated power but the lamp voltage is too high and the current is too low, it is being undercooled.

The fan control will keep the lamp in a free standing lamp housing at the correct temperature.

Recommended Procedure

1. If undercooled, check that there are no obstructions to airflow in front of the fan or in any related ducting.
2. If overcooled, look for external sources of cooling air (not controllable by the lamp housing fan) that could prevent the lamp housing circuitry from regulating the temperature.
3. Check that the fan control is working correctly. Refer to section IV.B, COOLING.

You cannot image the arc properly.

Normally you should be able to image the arc on a surface a few feet or meters away from the Lamp Housing by simply adjusting the condenser lens position. You should also be able to adjust the lamp position to center the beam in the condenser barrel output aperture.

Recommended Procedure:

1. Remove all power from the Lamp Housing and allow the lamp time to cool. Open the Lamp Housing and check the vertical position of the lamp. The arc gap should be positioned in the center of the reflecting mirror. If you have the appropriate adapters for your lamp you should be able to position it correctly, If not, call Oriel for further information.
2. Check for a broken condenser lens.

You cannot image the arc close to the Lamp Housing.

You should not have a problem imaging the arc some feet or meters away from the Lamp Housing. You may have problems imaging the arc close to the housing because the condenser lens adjustment does not allow the condenser to be located in an appropriate position. As you move the lens out, the image plane moves towards the lens. The closest image plane is determined by how far out you can move the lens.

Recommended Procedure

1. There are three tapped holes in the lens barrel to allow you to change the lens adjustment range. Remove the handle, carefully noting the location of each small part, and reassemble it with the screw in a different position to change the focus range. For reimaging close to the Lamp Housing, the screw should be in the hole closest to the front (output) of the lens barrel. Note that using a secondary focusing lens is often a better approach.

VI.3 INPUT CONNECTOR

Pin assignments for this connector are as follows:

- Pin A1 Power to the bottom of the lamp. Negative voltage for Xe and Hg(Xe) lamps; positive voltage for Hg lamps.
- Pin A2 Power to the top of the lamp. Positive voltage for Xe and Hg(Xe) lamps.; negative voltage for Hg lamps.
- Pin 1 Ground for lamp housing circuitry
- Pin 2 INTERLOCK (+). Connection to +12V required to enable Oriel power supply output. The lamp housing provides this connection through an interlock switch that is closed when the housing door is in place. Oriel power supply operation requires that both INTERLOCK (+) and INTERLOCK (-) be satisfied.
- Pin 3 Power (+12V) for lamp housing circuitry (including fan)
- Pin 4 Ignitor Drive. A ground level (nominal) that activates the lamp ignition circuit in an Oriel series 66900 lamp housing and the "boost" voltage in the power supply when the LAMP START button is pushed. Floating when button not pushed.
- Pin 5 Interlock (-). Connection to GND required to enable Oriel power supply output. The lamp housing provides this connection through an interlock switch that is closed when the housing door is in place. An overtemperature thermostat is included in the lamp housing circuit to break the interlock and disable the power supply if the housing temperature is too high. Oriel power supply operation requires that both INTERLOCK (+) and INTERLOCK (-) be satisfied.

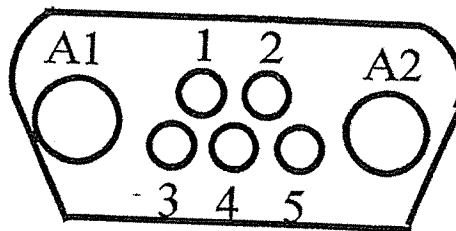


Figure 13: Lamp Housing Connector Pin Assignment

MATING CONNECTOR:

Body: ITT# DAM-7W2S-A197 (includes pins 1-5)
Pins: ITT# DM 53744-1 (requires 2 per connector, A1 and A2)
Backshell: standard 15-pin D-SUB

VII. DECLARATION OF CONFORMITY

Manufacturer's name: Newport Corporation
Manufacturer's address: 150 Long Beach Boulevard
Stratford, CT 06615
USA

Declares that the product:

Product Name A) Arc Lamp Housing

Model Number: 66900, 66901, 66902, 66903, 66904, 66905, 66912, 66701,
67003, 67005, 67006

**Conforms to the following
Product Specifications:**

Safety: EN 61010-1: 1993 + A2/ IEC 1010-1: 1990 + A1 + A2

EMC: EN 50081-1: 1992

EN 55022: 1994 / EN 55011: 1993 Class B

EN 50082-1: 1992

IEC 801-2:1991 / IEC 1000-4-2: 1995

ENV 50140: 1993 / EC 1000-4-3: 1995 / EN 61000-4-3: 1995

IEC 801-4: 1988 / IEC 1000-4-4: 1995

**Complies with the following
Directives:**

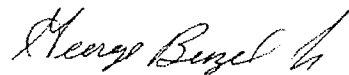
- the EMC Directive 89/336/EEC

- the Low Voltage Directive 73/23/EEC

**And accordingly, carries the CE
mark.**

A

Stratford, December, 1997
CT



(Signature)

George Buzel

(Name)

Director of Engineering

(Title)

VIII. WARRANTY AND RETURNS

Newport warrants that all goods described in this manual (except consumables such as lamps, bulbs, filters, ellipses, etc.) shall be free from defects in material and workmanship. Such defects become apparent within the following period:

1. All products described here, except spare parts: one (1) year or 3000 hours of operation, whichever comes first, after delivery of the goods to the buyer.
2. Spare parts: ninety (90) days after delivery of goods to the buyer.

Newport's liability under this warranty is limited to the adjustment, repair and/or replacement of the defective part(s). During the above listed warranty period, Newport shall provide all materials to accomplish the repaired adjustment, repair or replacement. Newport shall provide the labor required during the above listed warranty period to adjust, repair and/or replace the defective goods at no cost to the buyer ONLY IF the defective goods are returned, freight prepaid, to a Newport designated facility. If goods are not returned to Newport, and the user chooses to have repairs made at their premises, Newport shall provide labor for field adjustment, repair and/or replacement at prevailing rates for field service, on a portal-to-portal basis.

Newport shall be relieved of all obligations and liability under this warranty of:

1. The user operates the device with any accessory, equipment or part not specifically approved or manufactured or specified by Newport unless buyer furnishes reasonable evidence that such installations were not the cause of the defect. This provision shall not apply to any accessory, equipment or part which does not affect the safe operation of the device.
2. The goods are not operated or maintained in accordance with Newport's instructions and specifications.
3. The goods have been repaired, altered or modified by other than authorized Newport personnel.
4. Buyer does not return the defective goods, freight prepaid, to a Newport facility within the applicable warranty period.

IT IS EXPRESSLY AGREED THAT THIS WARRANTY SHALL REPLACE ALL WARRANTIES OF FITNESS AND MERCHANTABILITY. BUYER HEREBY WAIVES ALL OTHER WARRANTIES, GUARANTEES, CONDITIONS OR LIABILITIES, EXPRESSED OR IMPLIED, ARISING BY LAW OR OTHERWISE, WHETHER OR NOT OCCASIONED BY NEWPORT'S NEGLIGENCE.

This warranty shall not be extended, altered or varied except by a written document signed by both parties. If any portion of this agreement is invalidated, the remainder of the agreement shall remain in full force and effect.

CONSEQUENTIAL DAMAGES

Newport shall not be responsible for consequential damages resulting from misfunctions or malfunctions of the goods described in this manual. Newport's total responsibility is limited to repairing or replacing the malfunctioning or malfunctioning goods under the terms and conditions of the above described warranty.

INSURANCE

Persons receiving goods for demonstrations, demo loan, temporary use or in any manner in which title is not transferred from Newport, shall assume full responsibility for any and all damage while in their care, custody and control. If damage occurs, unrelated to the proper and warranted use and performance of the goods, recipient of the goods accepts full responsibility for restoring the goods to their condition upon original delivery, and for assuming all costs and charges.

RETURNS

Before returning equipment to Newport for repair, please call the Customer Service Department at (203) 377-8282. Have your purchase order number available before calling Newport. The Customer Service Representative will give you a Return Material Authorization number (RMA). Having an RMA will shorten the time required for repair, because it ensures that your equipment will be properly processed. Write the RMA on the returned equipment's box. Equipment returned without a RMA may be rejected by the Newport Receiving Department. Equipment returned under warranty will be returned with no charge for the repair or shipping. Newport will notify you of any repairs not covered by the warranty, with the cost of the repair, before starting the work.

Please return equipment in the original (or equivalent) packaging. You will be responsible for damage incurred from inadequate packaging, if the original packaging is not used.

Include the cables, connector caps and antistatic materials sent and/or used with the equipment, so that Newport can verify correct operation of these accessories.